
摄像机模型

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本课程三维重建篇所涉及的教学内容与课件参考了CS231A ,
感谢CS231A课程团队在课程建设方面所做的工作!

Machine Vision Technology							
Semantic information				Metric 3D information			
Pixels	Segments	Images	Videos	Camera		Multi-view Geometry	
Convolutions Edges & Fitting Local features Texture	Segmentation Clustering	Recognition Detection	Motion Tracking	Camera Model	Camera Calibration	Epipolar Geometry	SFM
10	4	4	2	2	2	2	2

摄像机几何

- 针孔模型 & 透镜
- 摄像机几何
- 其他摄像机模型

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摄像机几何

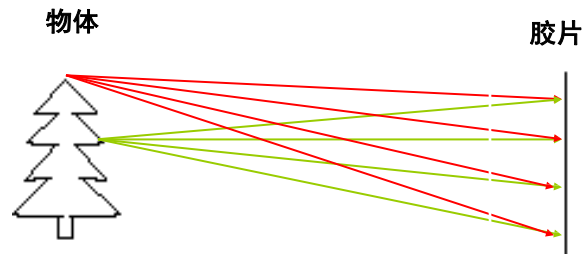
- 针孔模型 & 透镜
- 摄像机几何
- 其他摄像机模型

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我们如何记录世界？



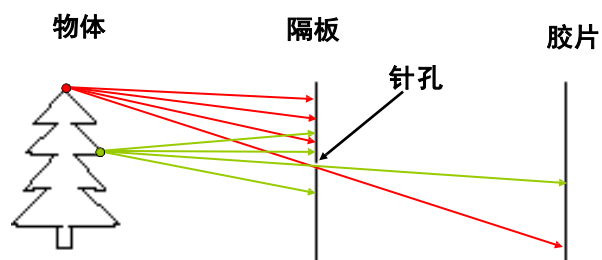
- 摄像机设计
 - 想法：将胶片直接放置在物体前方？

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针孔摄像机



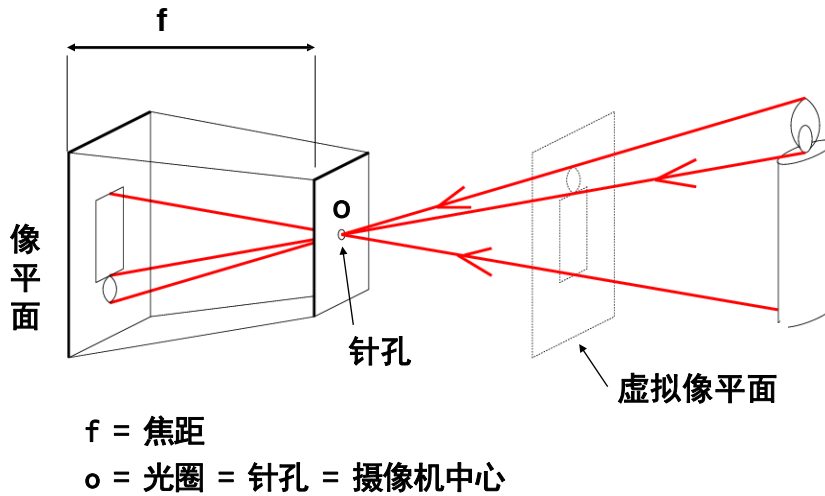
- 添加屏障——减少模糊

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针孔摄像机

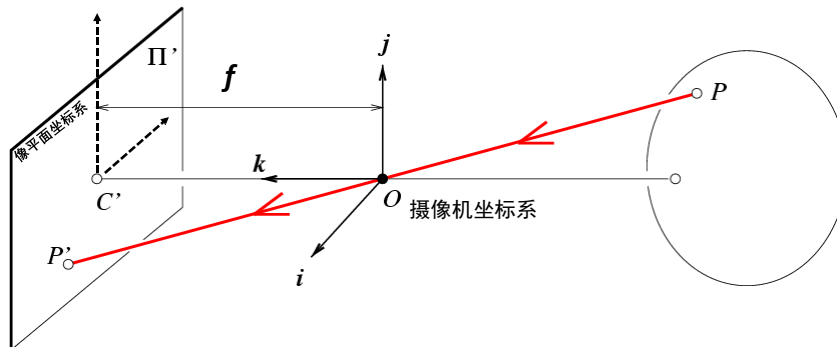


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针孔摄像机

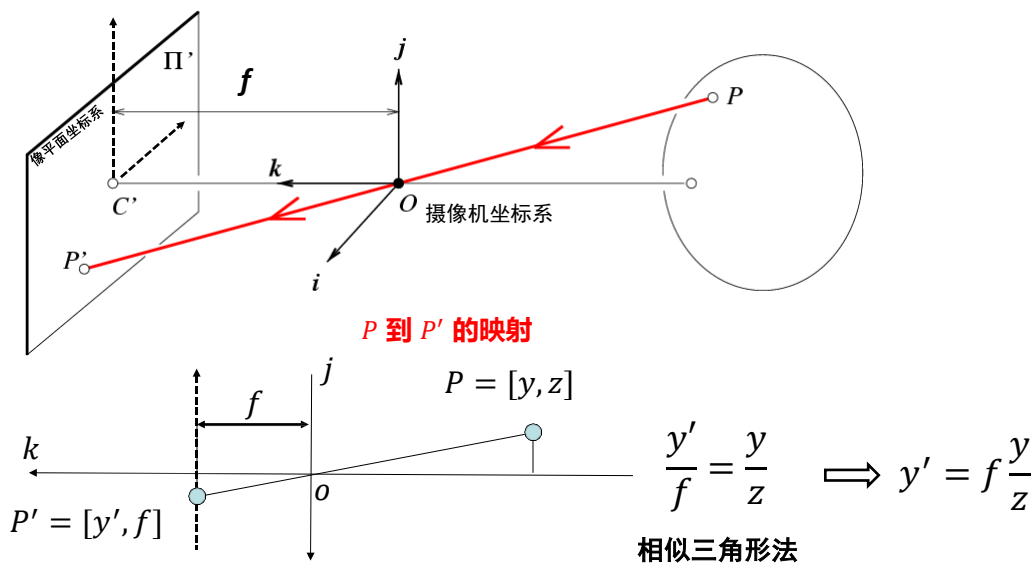


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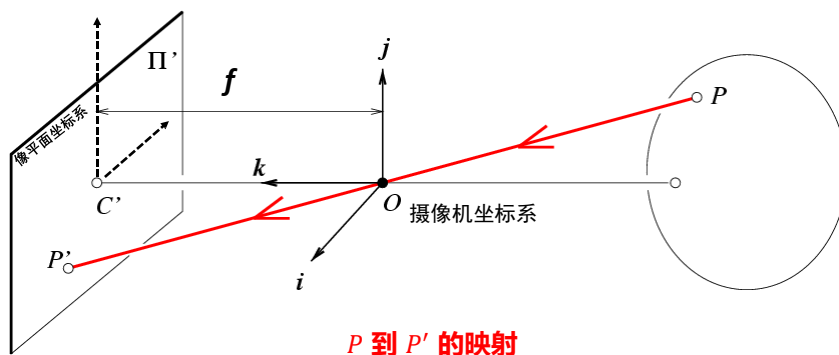


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针孔摄像机



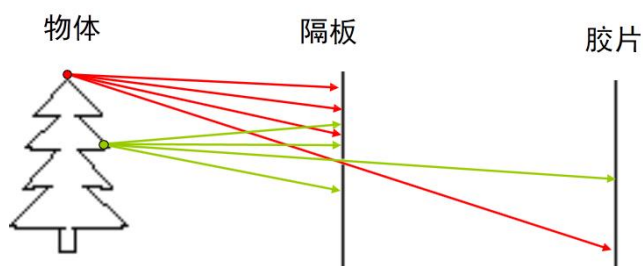
$$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow P' = \begin{bmatrix} x' \\ y' \end{bmatrix} \quad \begin{cases} x' = f \frac{x}{z} \\ y' = f \frac{y}{z} \end{cases}$$

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针孔摄像机



光圈的尺寸重要吗？



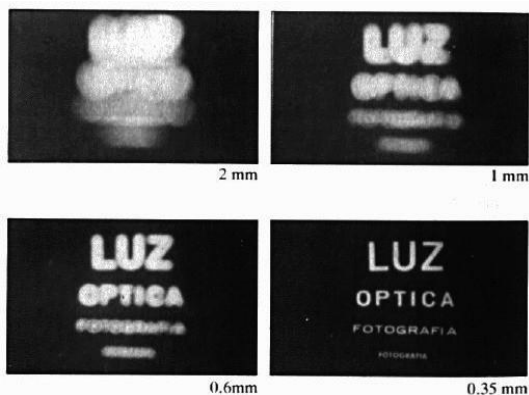
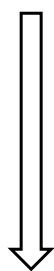
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针孔摄像机

缩小
光圈



随着光圈减小，成像效果如何变化？（越来越清晰、越来越暗）

如何应对到达胶片的光线变少？

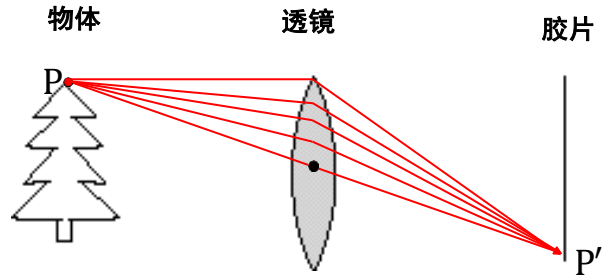
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摄像机 & 透镜

增加透镜!!!



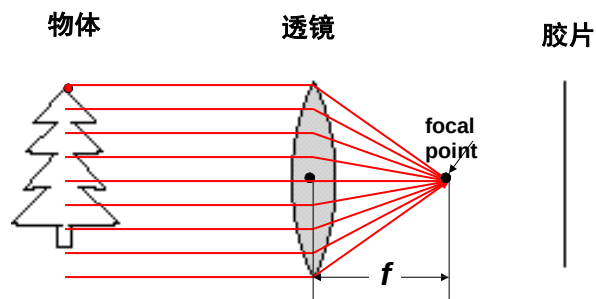
- 透镜将多条光线聚焦到胶片上，增加了照片的亮度

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摄像机 & 透镜



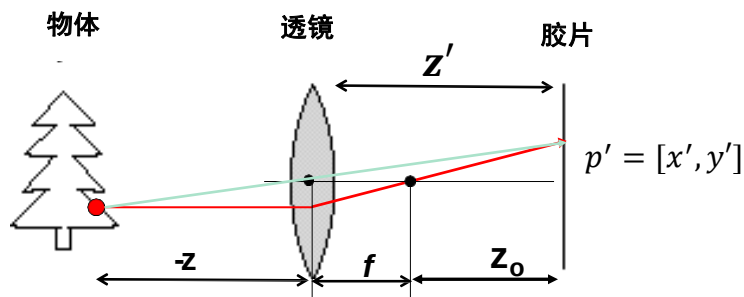
- 透镜将光线聚焦到胶片上
 - 所有平行于光轴的光线都会会聚到焦点，焦点到透镜中心的距离称为焦距。
 - 穿过中心的光线的方向不发生改变

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近轴折射模型



根据折射定律：

$$f = \frac{R}{2(n-1)}$$

$$z' = f + z_0$$

$$\begin{cases} x' = z' \frac{x}{z} \\ y' = z' \frac{y}{z} \end{cases}$$

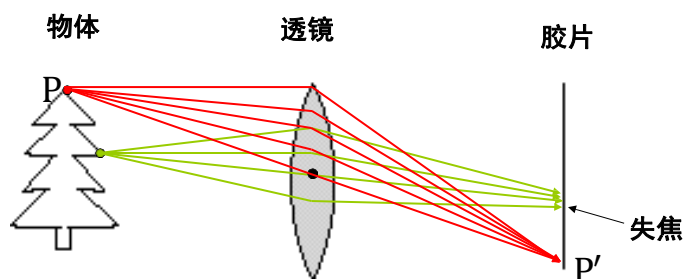
R 为透镜球面半径, n 为透镜折射系数

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透镜问题：失焦



• 透镜将光线聚焦到胶片上

- 物体“聚焦”有特定距离
- 景深

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透镜问题：失焦



微距摄像!!!

- 透镜将光线聚焦到胶片上
 - 物体“聚焦”有特定距离
 - 景深

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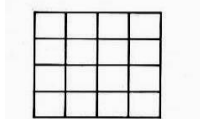
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透镜问题：径向畸变

- **径向畸变**: 图像像素点以畸变中心为中心点, 沿着径向产生的位置偏差, 从而导致图像中所成的像发生形变

没有畸变



枕形



畸变像点相对于理想像点沿径向向外偏移, 远离中心

桶形



畸变像点相对于理想点沿径向向中心靠拢



产生原因: 光线在远离透镜中心的地方比靠近中心的地方更加弯曲

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摄像机几何

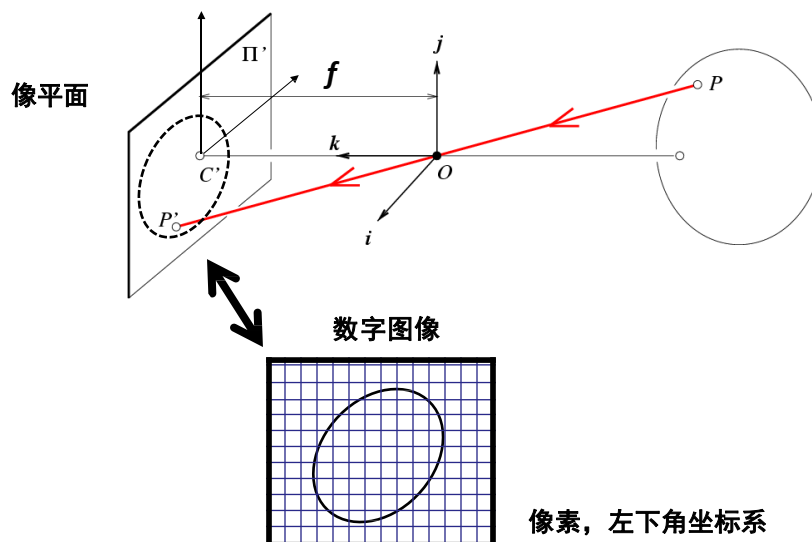
- 针孔模型 & 透镜
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像平面到像素平面

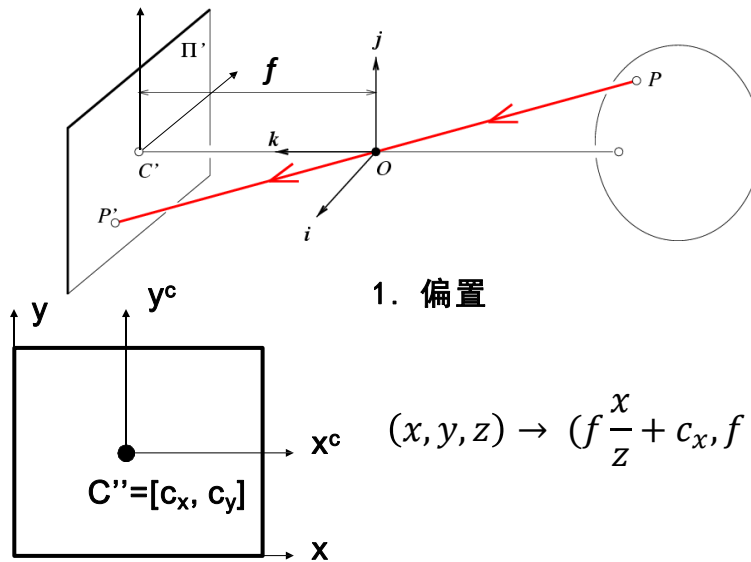


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像素坐标系

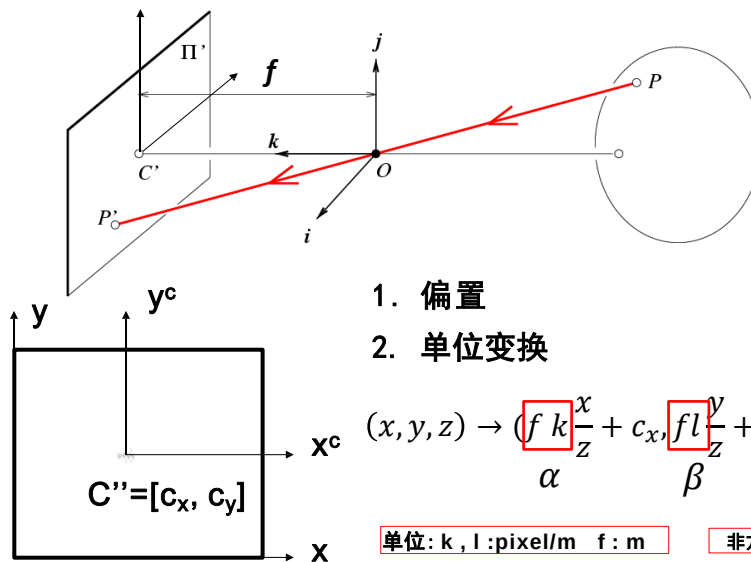


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像素坐标系

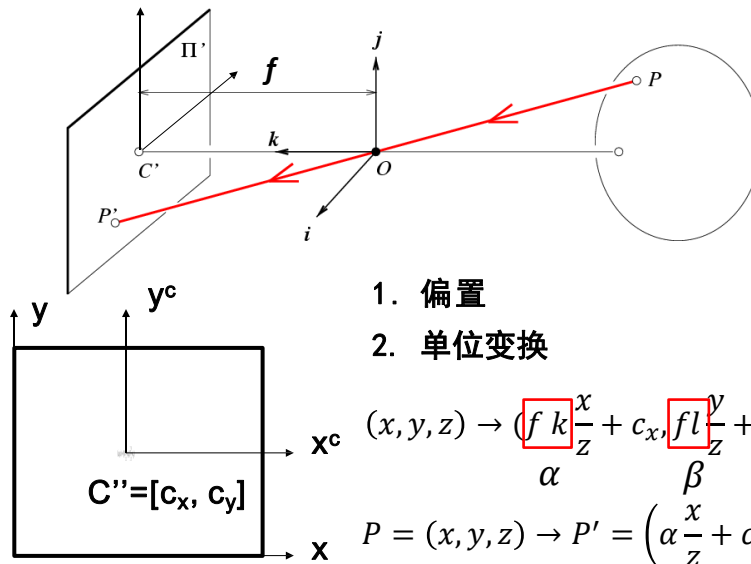


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像素坐标系



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问题: P 到 P' 的变换是线性的吗?

$$P = (x, y, z) \rightarrow P' = \left(\alpha \frac{x}{z} + c_x, \beta \frac{y}{z} + c_y \right)$$

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齐次坐标

E → H

$$(x, y) \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

图像点的齐次坐标

$$(x, y, z) \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

空间点的齐次坐标

H → E

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow (x/w, y/w)$$

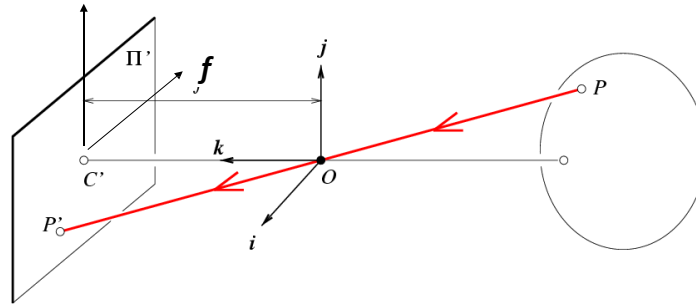
$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \Rightarrow (x/w, y/w, z/w)$$

齐次坐标系中的投影变换

$$P_h' = \begin{bmatrix} \alpha x + c_x z \\ \beta y + c_y z \\ z \end{bmatrix} = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \leftarrow P_h$$

$$\underbrace{P_h'}_{\text{齐次}} \rightarrow P' = \underbrace{\left(\alpha \frac{x}{z} + c_x, \beta \frac{y}{z} + c_y \right)}_{\text{欧式}}$$

摄像机的投影矩阵



$$P' = \begin{bmatrix} \alpha & 0 & c_x & 0 \\ 0 & \beta & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = MP$$

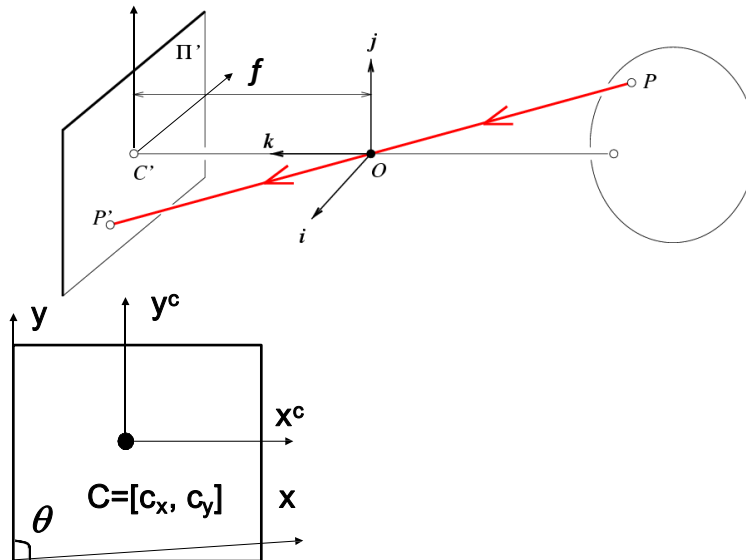
$$M = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x & 0 \\ 0 & \frac{\beta}{\sin \theta} & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

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摄像机偏斜

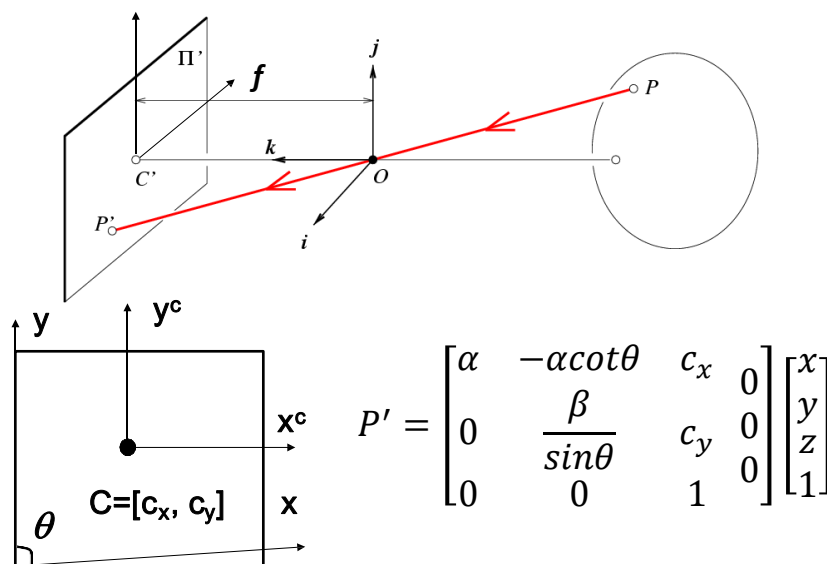


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摄像机偏斜

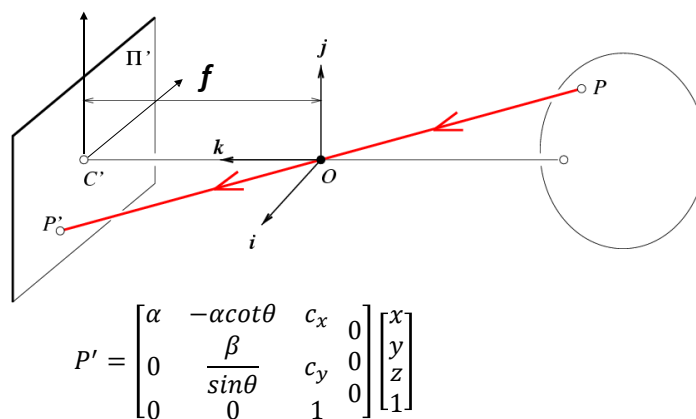


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摄像机坐标系下的摄像机模型



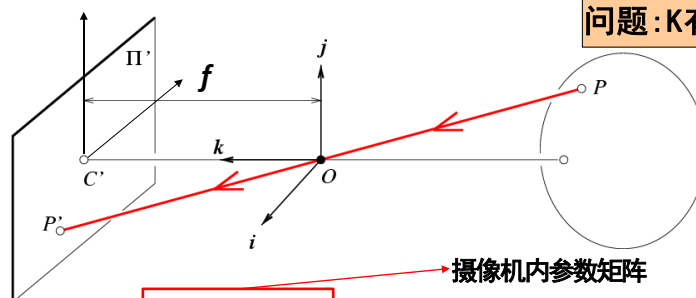
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摄像机坐标系下的摄像机模型

问题: K有多少个自由度?



摄像机内参数矩阵

$$P' = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x & 0 \\ 0 & \frac{\beta}{\sin \theta} & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

内参数决定了摄像机坐标系下空间点到图像点的映射!

$$K = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x \\ 0 & \frac{\beta}{\sin \theta} & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

$= M P$

投影矩阵

$= K [I \ 0] P$

$$M = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x & 0 \\ 0 & \frac{\beta}{\sin \theta} & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

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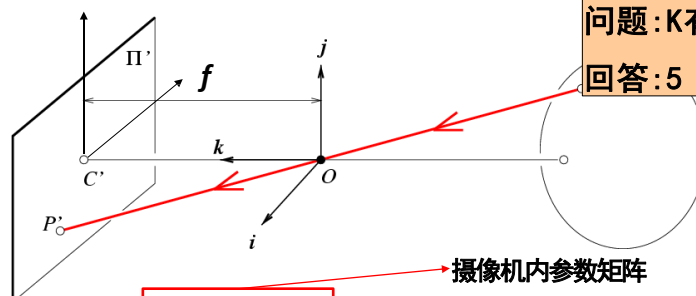
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摄像机坐标系下的摄像机模型

问题: K有多少个自由度?

回答: 5 DOF !



摄像机内参数矩阵

$$P' = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x & 0 \\ 0 & \frac{\beta}{\sin \theta} & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

内参数决定了摄像机坐标系下空间点到图像点的映射!

$$K = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x \\ 0 & \frac{\beta}{\sin \theta} & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

$= M P$

投影矩阵

$= K [I \ 0] P$

$$M = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x & 0 \\ 0 & \frac{\beta}{\sin \theta} & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

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规范化投影变换

$$P' = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_M \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \mathcal{R}^4 \xrightarrow{H} \mathcal{R}^3$$

$$P' = M P$$

$$P' \text{ 欧式坐标为 } \begin{bmatrix} \frac{x}{z} \\ \frac{y}{z} \\ 1 \end{bmatrix}$$

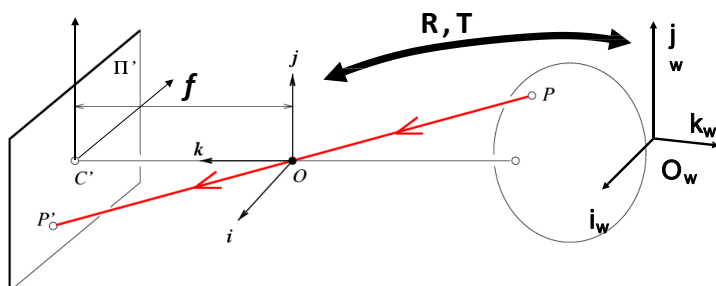
$$M = \begin{bmatrix} \alpha & -\alpha \cot \theta & c_x & 0 \\ 0 & \frac{\beta}{\sin \theta} & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

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世界坐标系



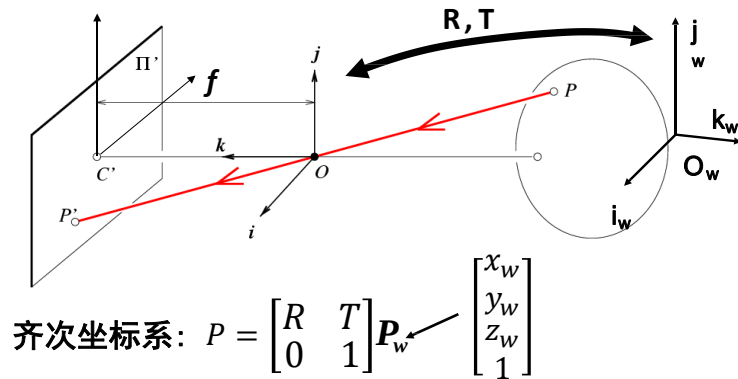
- 摄像机坐标系描述三维物体的空间信息是否方便？
- 如何将物体从世界坐标系转到摄像机坐标系？

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摄像机外参数

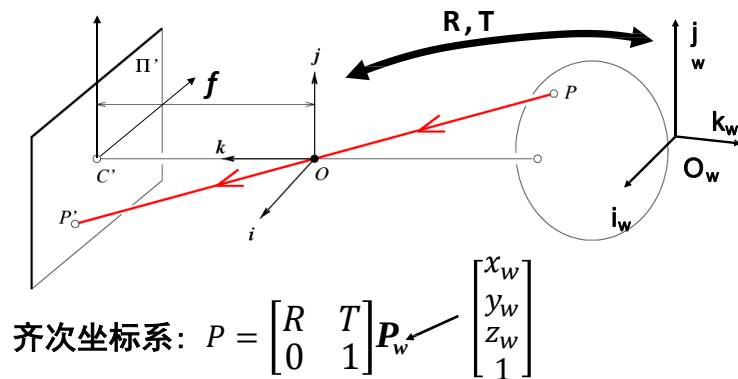


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摄像机外参数



外参数矩阵

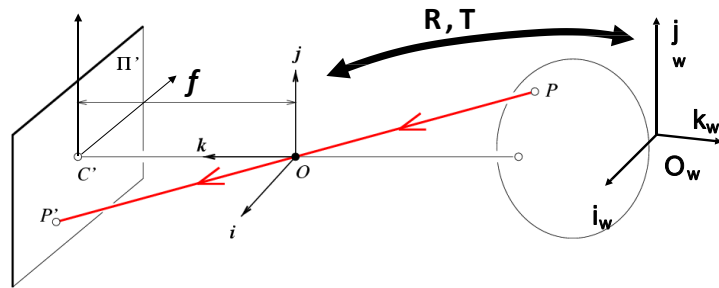
$$P' = K \begin{bmatrix} I & 0 \end{bmatrix} P = K \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} P_w = K \begin{bmatrix} R & T \end{bmatrix} P_w = M P_w$$

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摄像机几何



$$P' = K[I \ 0]P = K[I \ 0] \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} P_w = K \begin{bmatrix} R & T \end{bmatrix} P_w = MP_w$$

内部参数 外部参数

完整的摄像机模型！

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问题：各个符号的物理意义及其维度分别是什么？

$$P' = K[I \ 0]P = K[I \ 0] \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} P_w = K[R \ T]P_w = MP_w$$

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问题: 投影矩阵 M 有多少个自由度?

$$P' = K \begin{bmatrix} I & 0 \end{bmatrix} P = K \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} P_w = K \begin{bmatrix} R & T \end{bmatrix} P_w = M P_w$$

内部参数
外部参数

问题: P' 转换成欧式坐标该如何写?

$$P' = K \begin{bmatrix} I & 0 \end{bmatrix} P = K \begin{bmatrix} I & 0 \end{bmatrix} \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} P_w = K \begin{bmatrix} R & T \end{bmatrix} P_w = M P_w = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix} P_w$$

内部参数
外部参数

定理 (Faugeras, 1993)

$$M = K[R \ T] = [KR \ KT] = [A \ b] \quad A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

令 $M = (A \ b)$ 为 3×4 的矩阵, $a_i^T (i = 1, 2, 3)$ 表示由矩阵 A 的行

- M 是透视投影矩阵的一个充分必要条件是 $\text{Det}(A) \neq 0$
- M 是零倾斜透视投影矩阵的一个充分必要条件是 $\text{Det}(A) \neq 0$ 且

$$(a_1 \times a_3) \cdot (a_2 \times a_3) = 0$$

- M 是零倾斜且宽高比为1的透视投影矩阵的一个充分必要条件是 $\text{Det}(A) \neq 0$ 且

$$\begin{cases} (a_1 \times a_3) \cdot (a_2 \times a_3) = 0 \\ (a_1 \times a_3) \cdot (a_1 \times a_3) = (a_2 \times a_3) \cdot (a_2 \times a_3) \end{cases}$$

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投影变换的性质

1. 点投影为点
2. 线投影为线
3. 近大远小
4. 角度不再保持
5. 平行线相交



3D世界中的平行线在图像中相交于“影消点”

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摄像机几何

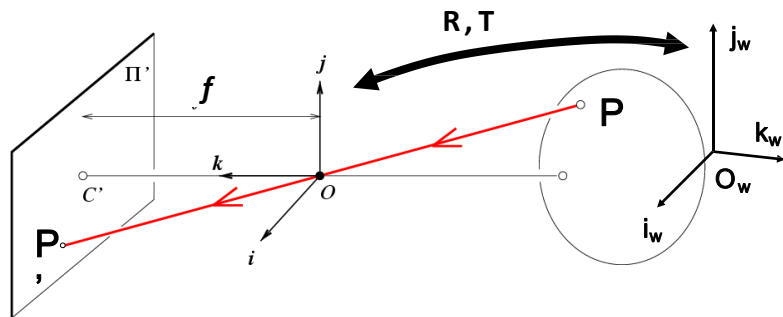
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- 摄像机几何
- 其他摄像机模型

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透视投影摄像机



$$P'_{3 \times 1} = MP_w = K_{3 \times 3} [R \quad T]_{3 \times 4} P_{w4 \times 1} \quad M = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix}$$

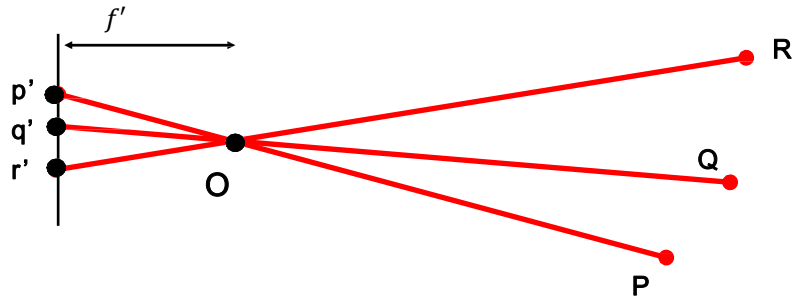
$$= \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix} P_w = \begin{bmatrix} m_1 P_w \\ m_2 P_w \\ m_3 P_w \end{bmatrix} \xrightarrow{E} \left(\frac{m_1 P_w}{m_3 P_w}, \frac{m_2 P_w}{m_3 P_w} \right)$$

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透视投影摄像机



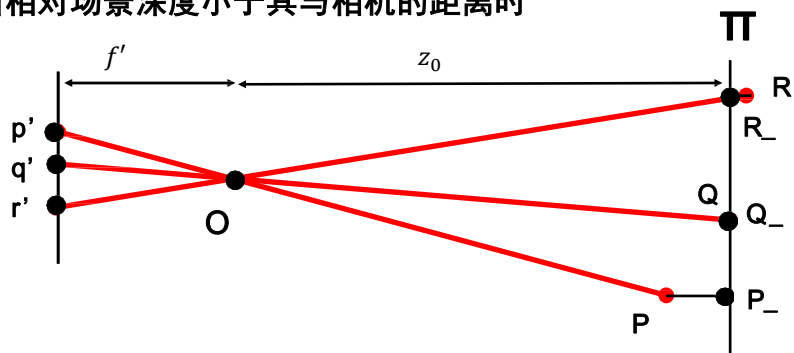
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弱透视投影摄像机

当相对场景深度小于其与相机的距离时

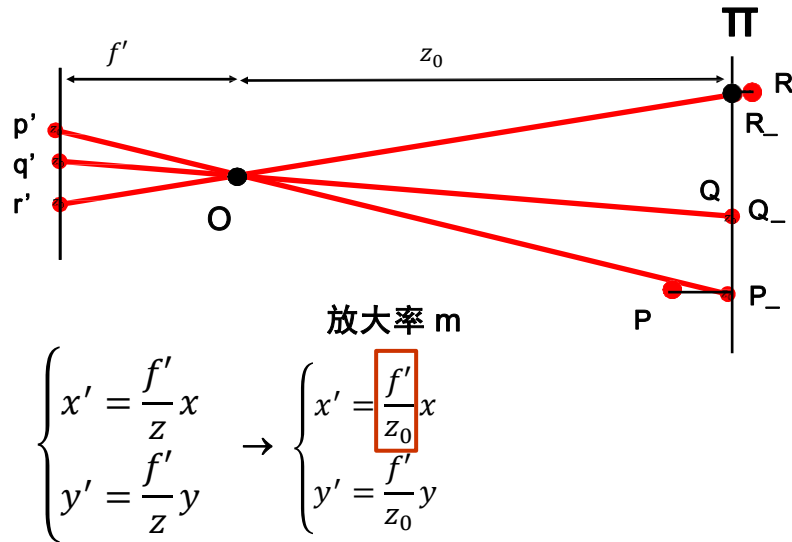


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弱透视投影摄像机

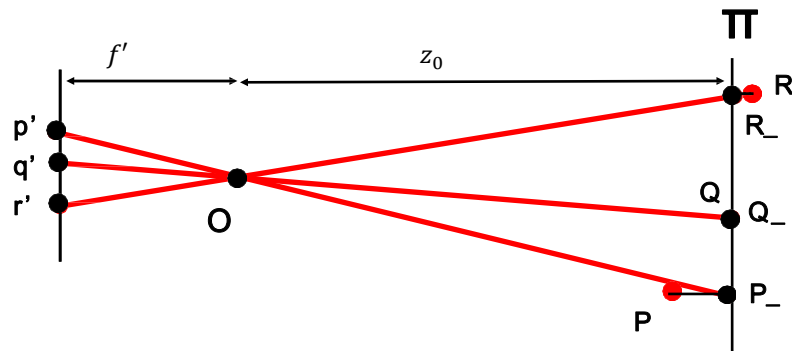


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弱透视投影摄像机



投影（透视）

弱透视

$$M = K[R \ T] = \begin{bmatrix} A_{2 \times 3} & b_{2 \times 1} \\ v_{1 \times 2} & 1 \end{bmatrix} \rightarrow M = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$$

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弱透视与透视投影摄像机

$$P' = MP_w = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix} P_w = \begin{bmatrix} m_1 P_w \\ m_2 P_w \\ m_3 P_w \end{bmatrix} \quad M = \begin{bmatrix} A & b \\ v & 1 \end{bmatrix} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix}$$

$$\xrightarrow{E} \left(\frac{m_1 P_w}{m_3 P_w}, \frac{m_2 P_w}{m_3 P_w} \right)$$

透视

$$P' = MP_w = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix} P_w = \begin{bmatrix} m_1 P_w \\ m_2 P_w \\ 1 \end{bmatrix} \quad M = \begin{bmatrix} A & b \\ v & 1 \end{bmatrix} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix}$$

$$\xrightarrow{E} (m_1 P_w, m_2 P_w)$$

↑ ↑
放 大 率

$$= \begin{bmatrix} m_1 & & & \\ & m_2 & & \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

弱透视

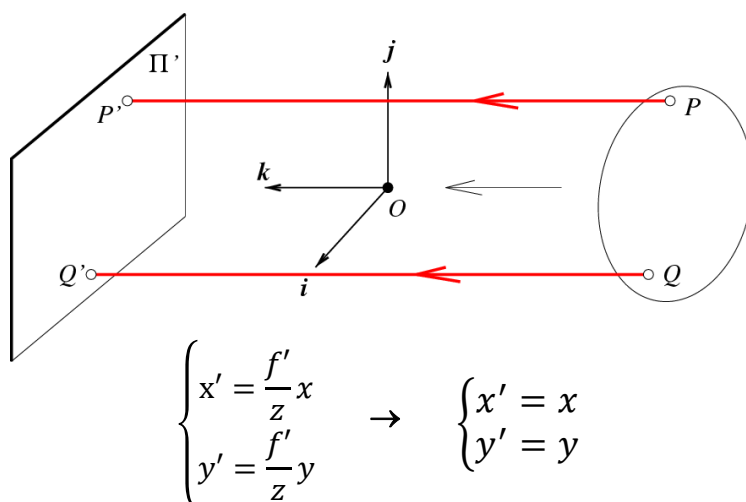
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正交投影摄像机

摄像机中心到像平面的距离无限远时



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各种摄像机模型的应用场合

- 正交投影
 - 更多应用在建筑设计 (AUTOCAD) 或者工业设计行业
- 弱透视投影在数学方面更简单
 - 当物体较小且较远时准确, 常用于图像识别任务
- 透视投影对于3D到2D映射的建模更为准确
 - 用于运动恢复结构或SLAM

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摄像机几何

- 针孔模型 & 透镜
- 摄像机几何
- 其他摄像机模型

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